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How Did South China Connect to and Separate From Gondwana? New Paleomagnetic Constraints From the Middle Devonian Red Beds in South China

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Abstract The paleogeographic relationship between South China and Gondwana is critical for understanding the dispersion of Gondwana, accretion of Asia, and evolution of the Paleo-Tethys. However, the lack of robust Devonian paleomagnetic data prevents a confirmative reconstruction of South China's connection to Gondwana and its subsequent separation during the Paleozoic. Here we report a new paleopole (33.6°N, 236.4°E; A_{95} = 3°) from the Givetian red beds (~385 Ma) in central South China. Fitting apparent polar wander paths between South China and Gondwana suggests that South China was connected to East Gondwana from the earliest Cambrian to Early Devonian, with its position closed to NW Australia. Thereafter, South China separated from Gondwana during ~400–385 Ma, as evidenced by their decoupled apparent polar wander paths. The paleomagnetic data suggest that the Paleo-Tethys Ocean between South China and East Gondwana had been up to ~1,600 km latitudinally wide by ~360 Ma.

Plain Language Summary South China is thought to have drifted away from the north Gondwana margin during the Paleozoic. Two critical questions remain unanswered: (1) How did South China connect to Gondwana? (2) When and how did South China break up from Gondwana? Here we present new paleomagnetic evidence for the connection and separation between South China and Gondwana. Our results, in combination with existing data, suggest that South China was connected to East Gondwana from ~540 to ~400 Ma, with its position closed to northwest Australia. South China broke up from Gondwana during ~400–385 Ma, accompanied with the opening of the Paleo-Tethys Ocean. By ~360 Ma, South China was separated from East Gondwana by the ocean with a N–S width of ~1,600 km.

Key Points:
- A new paleopole (33.6°N, 236.4°E; A_{95} = 3°) was obtained from the Givetian red beds (~385 Ma) in South China
- South China was connected to East Gondwana during ~540–400 Ma, with its position closed to northwest Australia
- South China broke up from East Gondwana during ~400–385 Ma and they were separated by an ocean with a N–S width of ~1,600 km by ~360 Ma

Supporting Information:
- Supporting Information S1
- Table S1

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1. Introduction

The South China Block (SCB) is considered as one of the continents that rifted off the margin of Gondwana, drifted northward, and accreted onto the East Asian continent from the Paleozoic to Mesozoic (Golonka et al., 2006; Metcalfe, 2013). Existing geologic, paleomagnetic, and paleontologic evidence suggests the affinity of the SCB with East Gondwana during the Early Paleozoic (Burrett et al., 1990, 2014; Han et al., 2015; Li et al., 2017; Nie, 1991; Zhang et al., 2001, 2015; Zhang, 2004). The presence of warm-water floras and absence of glacial deposits in Carboniferous–Permian strata of the SCB indicate no relationship with Gondwana (Metcalfe, 2013). However, the SCB's position relative to Gondwana during the Early Paleozoic remains highly controversial, with the SCB variously assigned to the proximity of NW India (Burrett et al., 2014; Li & Powell, 2001), NE India–W Australia (Cawood et al., 2013, 2018), or NW Australia (Han et al., 2015; Yang et al., 2004; Zhang et al., 2001), or as an isolated block away from Gondwana throughout the Early Paleozoic and Early Devonian (Cocks & Torsvik, 2013). The major cause of such uncertainty is the lack of robust paleomagnetic constraints on the position of the SCB during the Devonian, which is a key transitional period for the evolution of the Paleo-Tethys Ocean but has the least reliable data from the SCB and globally (Torsvik et al., 2012). In this study, we report a new paleopole that meets all seven criteria of Van der Voo (1990) from the Middle Devonian (Givetian) red beds in NW Hunan Province, China. In combination with available data, we present a new paleogeographic evolution model for the connection and separation between the SCB and Gondwana during the Paleozoic.

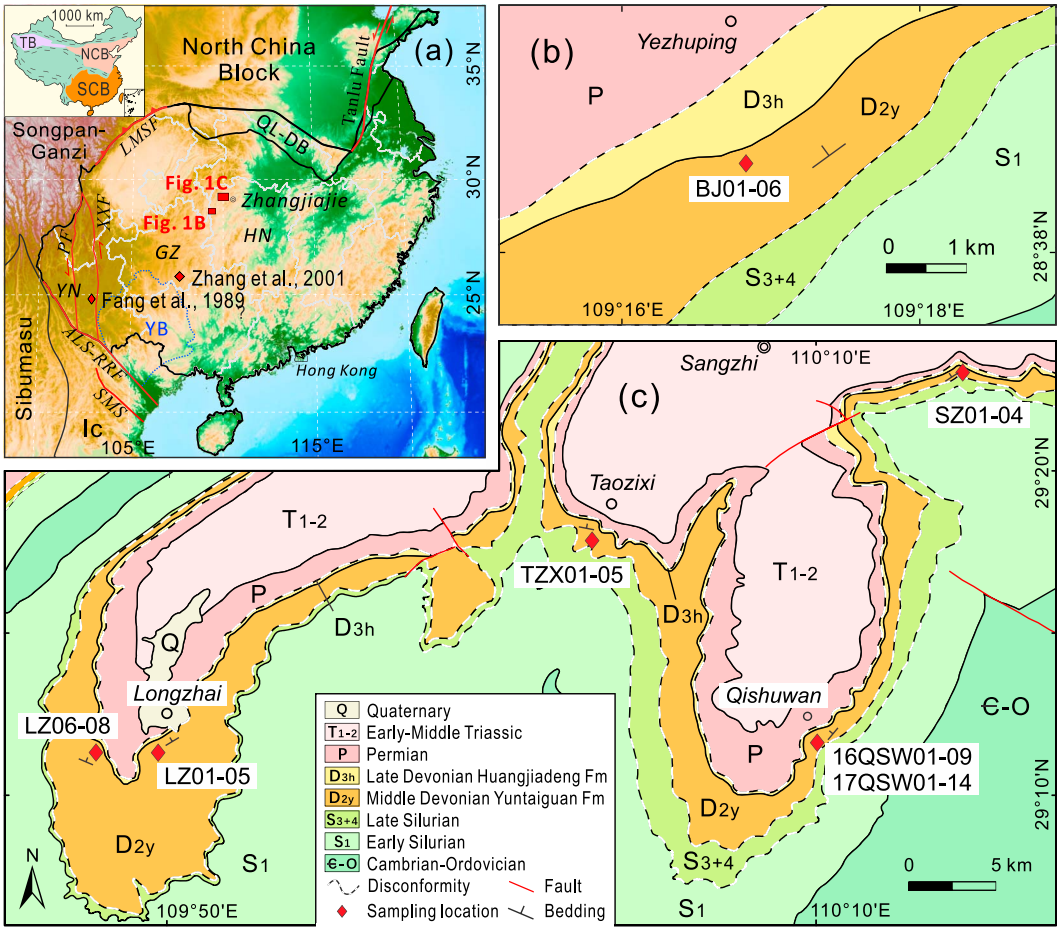


Figure 1. Geological setting and paleomagnetic sampling. (a) Tectonic outline of the South China Block (SCB) showing the research areas of this study in NW Hunan (HN). Previous research areas of Zhang et al. (2001) in SE Guizhou (GZ) and Fang et al. (1989) in East Yunnan (YN) are also marked. The blue dotted area shows the position of the Youjiang Basin (YB). The inset shows the tectonic outline of China with three major continental blocks. The topographic map was downloaded from (www.ngdc.noaa.gov). NCB = North China Block, QL-DB = Qinling–Dabie, TB = Tarim Block, Ic = Indochina, ALS-RRF = Ailaoshan–Red River Fault, SMS = Song Ma Suture, LMSF = Longmenshan Fault, PF = Panzhihua Fault, XXF = Xianshuihe–Xiaojiang Fault, Fm = Formation. Simplified geological maps of the (b) Baojing and (c) Zhangjiajie areas showing the locations of our paleomagnetic sampling sites.

2. Geological Setting and Paleomagnetic Sampling

Paleomagnetic samples were collected from the Zhangjiajie and Baojing regions in NW Hunan Province, South China (Figure 1a). In the study area, the Devonian strata disconformably overlie Upper-Silurian green siltstone and underlie Permian carbonate rocks (Figures 1b, 1c, and S1). The Devonian strata consist of the Middle Devonian Yuntaiguan Formation (YF) and Upper Devonian Huangjiadeng Formation (HF). The YF's lower part, 284 to 361 m thick, is characterized by reddish, medium- to thick-bedded quartz sandstone with thin interbeds of purple siltstone and silty shale. Its upper part, 69 to 242 m thick, is mainly composed of pink-white, thick-bedded quartz sandstone with minor purple-gray siltstone interbeds. The YF's age is assigned to the Givetian stage based on spore assemblages (Gao, 1989) and plant fossils (such as *Protolepidodendron scharyanum–Barrandeina dusliana*, *Protopteridium minutum*, *Lepidodendropsis* sp., and *Psilophytites* sp.; Chen, 1959; Zhu et al., 1983). The HF, 4 to 29 m thick, is characterized by Fe-bearing purple siltstone/mudstone or oolitic/pisolitic/brecciated hematite. Abundant plant fossils including *Leptophyloeum rhombicum* and *Cyclosligma kiliorkense* found in the HF denote its Late Devonian age (Hou et al., 1988). The Paleozoic strata were tilted due to the NW–SE compression of the Yanshanian (Late Jurassic–Early Cretaceous) tectonic event (Li et al., 2014; Liu et al., 2012), forming NNE trending folds with gentle bedding (Figures 1b and 1c).

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A total of 400 samples of 46 sites were collected from five sections of the YF (Figures 1 and S1 and S2 in the supporting information). Among these, 23 sites were arranged continuously covering an ~370-m-thick interval in the Qishuwan Section, in which the YF is up to 600 m thick and exposed continuously from its bottom to top. The other 23 sites covered the other four sections (Baojing, Longzhai, Sangzhi, and Taozixi) with different bedding attitudes. Most of the samples are purple siltstone, and a few are quartz sandstone. Each site contains 6–15 samples, spanning a stratigraphic thickness of 0.5–15 m. All the sample cores were oriented with a magnetic compass and Sun compass when possible, and the differences were less than 1° between the two methods, which indicates insignificant local magnetic anomaly.

3. Methods and Results

Anisotropy of magnetic susceptibility was measured first using an AGICO MFK1 kappa-bridge. After that, remanent magnetizations were measured on a 2G-755K three-axis cryogenic magnetometer, and thermal demagnetizations were performed on an MMTDSC furnace, both of which have internal residual fields of less than 10 nT. To further characterize the magnetic mineralogy, representative samples were chosen for isothermal remanent magnetization (IRM) acquisition curves, backfield demagnetizations of IRM, and Lowrie test (Lowrie, 1990). Remanent magnetization measurements and demagnetizations were conducted in a magnetically shielded room with a residual field of less than 300 nT in the Paleomagnetism and Environmental Magnetism Laboratory at the China University of Geosciences, Beijing. Magnetic components were determined using principal component analysis (Kirschvink, 1980) with maximum angle deviation less than 15°, and mean directions were calculated using Fisher statistics (Fisher, 1953). Paleomagnetic data were analyzed using the software packages of PaleoMag (Jones, 2002), PaleoMac (Cogné, 2003), and PmagPy (Tauxe et al., 2016). Plate reconstructions were built using the open-source software GPlates (www.gplates.org).

Anisotropy of magnetic susceptibility measurements revealed an oblate anisotropy of magnetic susceptibility ellipsoid (shape parameter $T = 0.7$), and in stratigraphic coordinates, the minimum susceptibility axes (K_3) are mainly perpendicular to the bedding and the maximum, whereas the medium axes (K_1 and K_2) are scattered within the bedding plane (Figure S3), indicating a normal magnetic fabric of sedimentary rocks. Stepwise thermal demagnetization of all the samples produced consistent results among the five sections. The natural remanent magnetization intensities range from 1 to 20 mA/m. Most samples recorded two well-defined components (Figures 2a, 2b, and S4a–S4j). A low-temperature component removed below 400 °C resembles the present geomagnetic field direction in geographic coordinates (Figure S5) and is interpreted as the viscous remanent magnetizations acquired in the present geomagnetic field. A high-temperature component (HTC) was identified between 400 °C and 690 °C, and it decayed linearly toward the origin in orthogonal vector diagrams. Magnetizations decayed abruptly to zero at unblocking temperatures of 660–690 °C (Figure S4k), indicating that the HTC was carried by hematite, as also supported by IRM acquisition curves, backfield demagnetization of IRM, and the Lowrie test (Figure S6 and Text S1), which revealed the dominance of high-coercivity and high unblocking temperature magnetic minerals (Dunlop & Özdemir, 1997).

The HTC of the Qishuwan Section has dual-polarity directions: mostly southwest and down, and minor northeast and up (Figures 2, S1, and S2a). The magnetic polarity was determined based on the Carboniferous and younger poles from the SCB (Zhang et al., 2015). The NE upward (SW downward) directions are suggested as geomagnetically normal (reversal) polarity. The HTC yielded a site-mean direction of $Dg = 36.5°$, $Ig = -11.6°$ ($k = 44$, $\alpha_{95} = 3.8°$, $N = 34$) in situ and $Ds = 42.7°$, $Is = -18.1°$ ($k = 58.1$, $\alpha_{95} = 3.3°$) after tilt correction (Figures 2c and 2d). The HTC passes a positive fold test (McFadden, 1990) for both "Xi 1" and "Xi 2" tests at the 95% and 99% confidence levels (critical "Xi" 95% (99%) = 6.783 (9.590); "Xi 1" IS = 12.19, "Xi 1" TC = 0.9502; "Xi 2" IS = 18.18, "Xi 2" TC = 3.358; Table S1), a positive bootstrap fold test (Figure S7; Tauxe & Watson, 1994; Tauxe et al., 2016), and a C-class reversal test (angle between the two averages of gamma (γ_o) = 4.8° < critical gamma (γ_c) = 11.8°; Table S1; McFadden & McElhinny, 1990), demonstrating a primary magnetization. The HTC does not show obvious elongated distribution, and elongation/inclination correction (Figure S8; Tauxe & Kent, 2004) was performed on the HTC of the Qishuwan Section. It yielded a flattening factor (f) of 0.8, corresponding to a corrected direction of $Ds' = 42.9°$, $Is' = -21.9°$ ($k = 54.7$, $\alpha_{95} = 3.4°$, $N = 34$) and suggesting insignificant shallowing; thus, we use the

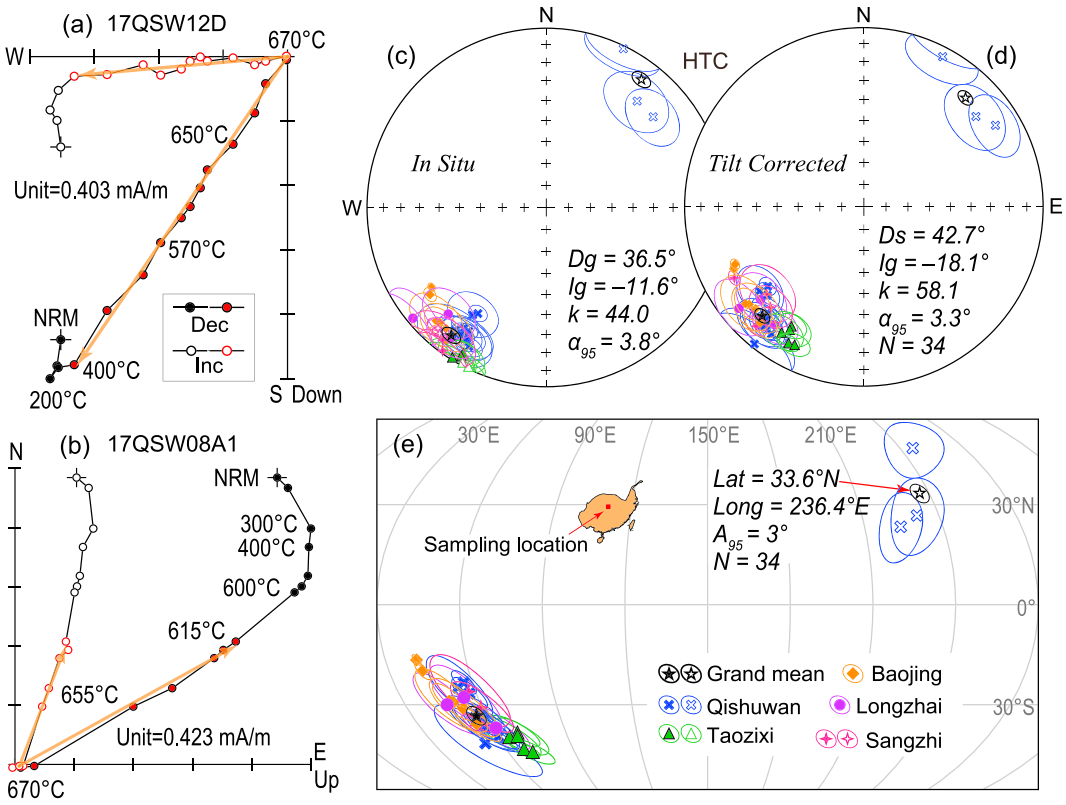


Figure 2. Paleomagnetic results of the Yuntaiguan Formation from the SCB. Orthogonal projection diagrams (Zijderveld, 1967) for representative samples of (a) reversal and (b) normal polarities in geographic coordinates. Red open/solid circles are demagnetization steps used in least squares fits (Kirschvink, 1980) of the high-temperature components (HTC) shown as orange vectors. Dec = declination, Inc = inclination, NRM = natural remnant magnetization. (c) In situ and (d) tilt-corrected site-mean directions in equal-area plots. (e) Mollweide projection for 34 site VGPs from the five sections and their grand mean pole in stratigraphic coordinates. Solid and open symbols represent VGPs derived from reversal and normal polarity directions, respectively. The symbols in (c), (d), and (e) are consistent.

uncorrected direction. The paleopole was calculated based on virtual geomagnetic poles (VGPs) at the site level (Figure 2e) and section level. Little difference was observed between the two methods (Table S1). We choose the site-VGP level pole (33.6°N, 236.4°E; A_{95} = 3°, N = 34) to reduce any potential influence from secular variation of the stratigraphically thin sections. The pole diverges significantly from all younger poles from the SCB (Figure 3a), further confirming its primary origin. In summary, the pole meets all seven reliability criteria of Van der Voo (1990) (Text S2) and represents a key pole for the SCB in the Givetian stage (~385 Ma).

Previously published Devonian paleomagnetic data sets from East Yunnan (Fang et al., 1989) and Guizhou (Zhang et al., 2001) are consistent in low inclination and the presence of dual-polarity directions, but there is ~50° difference in declination that cannot be neglected. This difference is probably due to regional rotation of Eastern Yunnan, which is bounded by the Panzhihua and Xianshuihe–Xiaojiang Faults (Figure 1a; Yang & Besse, 1993). Our new results are consistent with but slightly different from the Early–Middle Devonian results of Zhang et al. (2001). The consistency attests to the reliability of those data from Guizhou, whereas the slight difference in directions may have resulted from the disparity in age. Thus, in the following discussions, the pole of Fang et al. (1989) is not used, whereas the two poles of Zhang et al. (2001) derived from late Early Devonian–early Middle Devonian and Late Devonian–Early Carboniferous strata are taken as reference poles at ~400 and ~360 Ma for the SCB, respectively.

4. Paleogeographic Implications

A new model of paleogeographic evolution was obtained by fitting the apparent polar wander paths (APWPs) between the SCB and Gondwana (Figure 3). The SCB APWP was mainly modified from Zhang et al. (2015),

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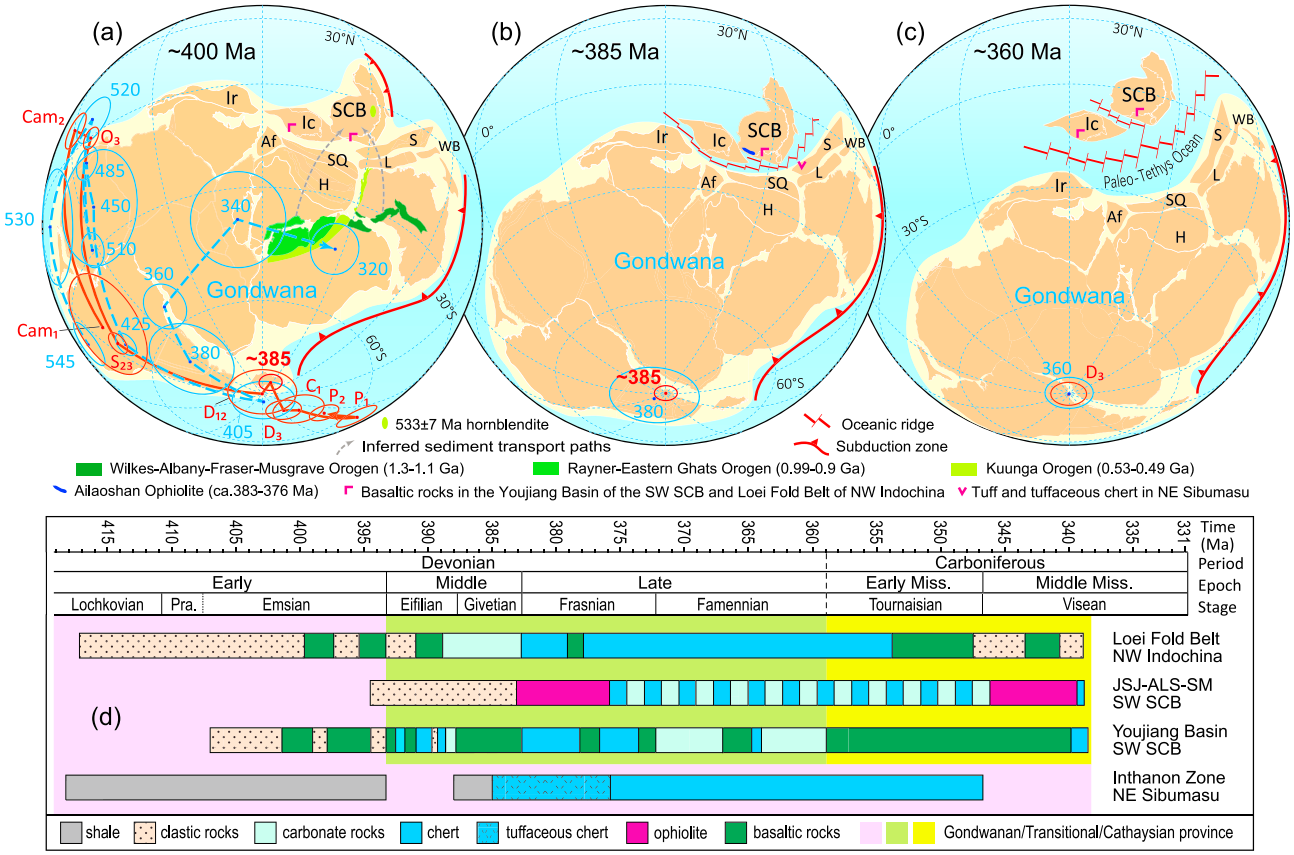


Figure 3. Paleogeographic reconstructions at (a) ~400 Ma, (b) ~385 Ma, and (c) ~360 Ma. The fit of Gondwana is mainly modified after Torsvik et al. (2012), except for the relative position of Madagascar–India–Sri Lanka to Africa and Antarctica, which is adopted from Reeves et al. (2016) as a tight reconstruction based on the updated ocean-floor fracture zone data of the Indian Ocean. The poles and labels of Gondwana are colored blue. The SCB poles are marked as orange circles with red labels. The ages of all number-labeled paleopoles are in Ma. Orogenic belts in (a), including the Wilkes–Albany–Fraser–Musgrave (1.3–1.1 Ga), Rayner–Eastern Ghats (0.99–0.9 Ga), and Kuunga (0.53–0.49 Ga) orogens, are modified from Boger (2011) and Cawood et al. (2018). Af = Afghanistan, Ic = Indochina, H = Himalaya, Ir = Iran, L = Lhasa, S = Sibumasu, SQ = South Qiangtang, WB = West Borneo. Cam_1/Cam_2 = early/middle Cambrian, O_3 = Late Ordovician, S_{23} = Middle–Late Silurian, D_{12} = Early–Middle Devonian, D_3 = Late Devonian, C_1 = Early Carboniferous, P_1/P_2 = Early/Late Permian. (d) Simplified Devonian–Early Carboniferous rock records and biotic provinces for the separation process between the SCB and neighboring blocks. Data sources: the time scale was created using TSCreator (https://timescalecreator.org/index/index.php; Ogg et al., 2016), the Loei Fold Belt was modified from Udchachon et al. (2017), JSJ-ALS-SM (Jinshajiang–Ailaoshan–Song Ma) from Jian et al. (2009) and Lai et al. (2014), the Youjiang Basin from Shi et al. (2006), the Inthanon Zone from Hara et al. (2010), and biotic provinces from Metcalfe (2013).

combined with the Late Ordovician pole (Han et al., 2015) and our new Givetian pole (Table S2). The Gondwana APWP is adopted from Li and Powell (2001) and McElhinny et al. (2003) (Table S3). By rotating the SCB to South Africa using Euler rotation (54°N, −12°E, −32°), four robust SCB poles of early Cambrian (~540 Ma; Lin et al., 1985), middle Cambrian (~520 Ma; Bai et al., 1998; Yang et al., 2004), Middle Silurian (~425 Ma; Huang et al., 2000; Opdyke et al., 1987), and Early Devonian (~400 Ma; Zhang et al., 2001) ages match well with coeval Gondwana poles. The Late Ordovician poles show overall consistency, although they are not well defined in both the SCB and Gondwana (Figure 3a). The agreement of APWPs suggests that the SCB was connected to East Gondwana from the earliest Cambrian to Early Devonian (~540–400 Ma), with its SW margin facing NE India–NW Australia.

Geological evidence supporting this reconstruction includes (1) Early Paleozoic strata in the SCB contain abundant 1.1–1.15- and 0.9–0.96-Ga detrital zircons that have no local sources in the SCB and were likely derived from the Wilkes–Albany–Fraser Belt between SW Australia–Antarctica and the Rayner–Eastern Ghats Belt between East Antarctica and India (Figure 3a), respectively (Cawood et al., 2018); (2) an early Cambrian (~533 Ma) metamorphic event recorded in hornblendite was discovered in Cathaysia and interpreted as an extension of the Kuunga Orogeny between eastern India and Australia–East Antarctica

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(Figure 3a; Li et al., 2017); and (3) paleontological evidence suggests the proximity of the SCB to both India and Australia during the Early Paleozoic (Figure 3d; Metcalfe, 2013; Wang et al., 2010).

Our model also permits reconstructions for some small blocks of SE Asia between the SCB and Gondwana as supported by recently published data (Figure 3a). Based on animal fossil records (Burrett et al., 1990), tectono-stratigraphic correlation (Zhu et al., 2013), and provenance analyses (Burrett et al., 2014; Song et al., 2017; Zhu et al., 2011), Himalaya and South Qiangtang were adjacent to North India, whereas Lhasa and Sibumasu were located next to NW Australia. The position of Sibumasu outboard NW Australia and close to Lhasa is also supported by paleomagnetic evidence from the Woniusi basalt (Xu et al., 2015). Indochina was located between the west SCB and South Qiangtang, based on the findings that they share correlatable sedimentological and animal fossil records (Cocks & Torsvik, 2013; Thanh et al., 1996; Wang et al., 2010), and similar detrital zircon age spectra (Usuki et al., 2013).

The decoupled APWPs of the SCB and Gondwana suggest that the SCB began to break up from Gondwana after ~400 Ma. The Devonian–Carboniferous segment of the Gondwana APWP displays a great loop, whereas the Middle Devonian–Permian poles of the SCB show the opposite direction (Figure 3a; Huang et al., 2008; Wu et al., 1999; Zhang et al., 2001). The Emsian basaltic rocks in the Youjiang Basin of the southwest SCB (Zhao et al., 1996) and the Loei Fold Belt of NW Indochina (Udchachon et al., 2017) could represent evidence of initial rifting (Figures 3a and 3d). The rift–drift process is inferred to have occurred during the Emsian–Givetian, leading to the incipient spreading of the eastern Paleo-Tethys Ocean between the Cimmerian continent (e.g., Sibumasu and South Qiangtang) and its northern blocks (e.g., South China and Indochina) (Metcalfe, 2013). However, we estimate that the SCB did not drift far from Gondwana at ~385 Ma, as their poles at this time still closely matched (Figures 3a and 3b). After the Givetian, as East Gondwana migrated toward the high-latitude region, the SCB remained in the equatorial zone, and the Paleo-Tethys Ocean between them spread quickly, accompanied with enhanced volcanic activity and extensive chert developing in the southwest SCB (Zhao et al., 1996) and NE Sibumasu (Hara et al., 2010). By the end of the Devonian (Figure 3c), the SCB was separated from East Gondwana by the Paleo-Tethys Ocean whose latitudinal width was ~1,600 km estimated based on the paleomagnetic data, resulting in dramatic differences in the sedimentary and paleontological records between the SCB and Gondwanan blocks (as recorded in NE Sibumasu and Australia; Hara et al., 2010; Metcalfe, 2013).

Between the SCB and Indochina, existing fish fossils suggest their connection until the Mid-Devonian (Thanh et al., 1996), whereas later ophiolites in the Ailaoshan (~383–376 Ma) and Jinshajiang (~346–341 Ma) sutures indicate the spreading of the Ailaoshan Ocean from the Late Devonian to Early Carboniferous (Figures 3b–3d; Jian et al., 2009). These findings, combined with the Late Permian subduction and Triassic collision (Cai & Zhang, 2009), imply the evolution of a complete Wilson cycle between these two blocks.

The separation processes between the SCB, Indochina, and Gondwana are also consistent with paleobiogeographic evidence that the biotic province of the SCB and Indochina changed from the Gondwanan biotic province (Early Devonian) to a transitional province (Middle–Late Devonian), and then to the Cathaysian province (Carboniferous–Permian), whereas Sibumasu remained in the Gondwanan province until the Early Permian (Figure 3d; Metcalfe, 2013).

5. Conclusions

A new paleomagnetic pole of 33.6°N, 236.4°E (A_{95} = 3°, N = 34) was obtained from the red beds of the Givetian Yuntaiguan Formation in the central SCB. The pole passes a C-class reversal test and a fold test, and differs from all younger poles of the SCB, meeting all seven of Van der Voo's criteria and manifesting its primary origin. This new paleopole, in combination with existing robust poles, locates the SCB in the equatorial region throughout the Paleozoic. Comparison of the Paleozoic APWPs between the SCB and Gondwana suggests that the SCB was connected to East Gondwana from the earliest Cambrian to Early Devonian, with its position closed to NW Australia. The SCB began to break up from Gondwana during ~400–385 Ma, accompanied with the opening of the Paleo-Tethys Ocean, but it was not far away from Gondwana until ~385 Ma. During ~385–360 Ma, as East Gondwana drifted toward the southern polar region, the SCB remained in the equatorial zone, and was separated from Gondwana by the Paleo-Tethys Ocean with a latitudinal width of ~1,600 km by ~360 Ma.

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